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Cumulative exposure to the estimated glucose disposal rate and incident stroke in individuals with cardiovascular–kidney–metabolic syndrome stages 0–4: 6-year longitudinal evidence from CHARLS

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Abstract

Background The estimated glucose disposal rate (eGDR), an established measure of peripheral insulin sensitivity, contributes to stratifying the risk of cardio-cerebrovascular events. Nevertheless, the association between long-term eGDR exposure and stroke incidence throughout all stages (0–4) of cardiovascular–kidney–metabolic (CKM) syndrome remains unknown.

Methods A cohort of 5248 individuals was drawn from the China Health and Retirement Longitudinal Study (CHARLS). For each participant, eGDR values for the years 2012 and 2015 were calculated using the equation: $21.158 - [0.090 \times \text{WC (cm)}] - [3.407 \times \text{HTN (presence = 1)}] - [0.551 \times \text{HbA1c (\%)}]$. Cumulative eGDR was calculated as $(\text{eGDR}_{2012} + \text{eGDR}_{2015})/2 \times \text{time (2015–2012)}$. K-means clustering was used to analyse eGDR values from both 2012 and 2015 to identify distinct change patterns. To assess associations with stroke risk, we utilised multivariable logistic regression and restricted cubic spline models.

Results During the 2015–2018 follow-up period, a total of 336 incident stroke cases were documented. Four distinct eGDR change patterns were identified. In fully adjusted models, compared with the participants in the persistent low pattern (Class 2), those in the moderate–high stable (OR 0.43, 95% CI: 0.31–0.58), stable high (OR 0.29, 0.19–0.43), and rapid decrease (OR 0.66, 0.47–0.91) patterns exhibited significantly lower stroke risk. Furthermore, each 1-unit increase in cumulative eGDR was associated with a 5% reduction in stroke odds (OR 0.95, 0.93–0.96). Restricted cubic spline analysis confirmed a linear inverse relationship between cumulative eGDR and stroke risk ($P < 0.001$; P for nonlinearity = 0.259).

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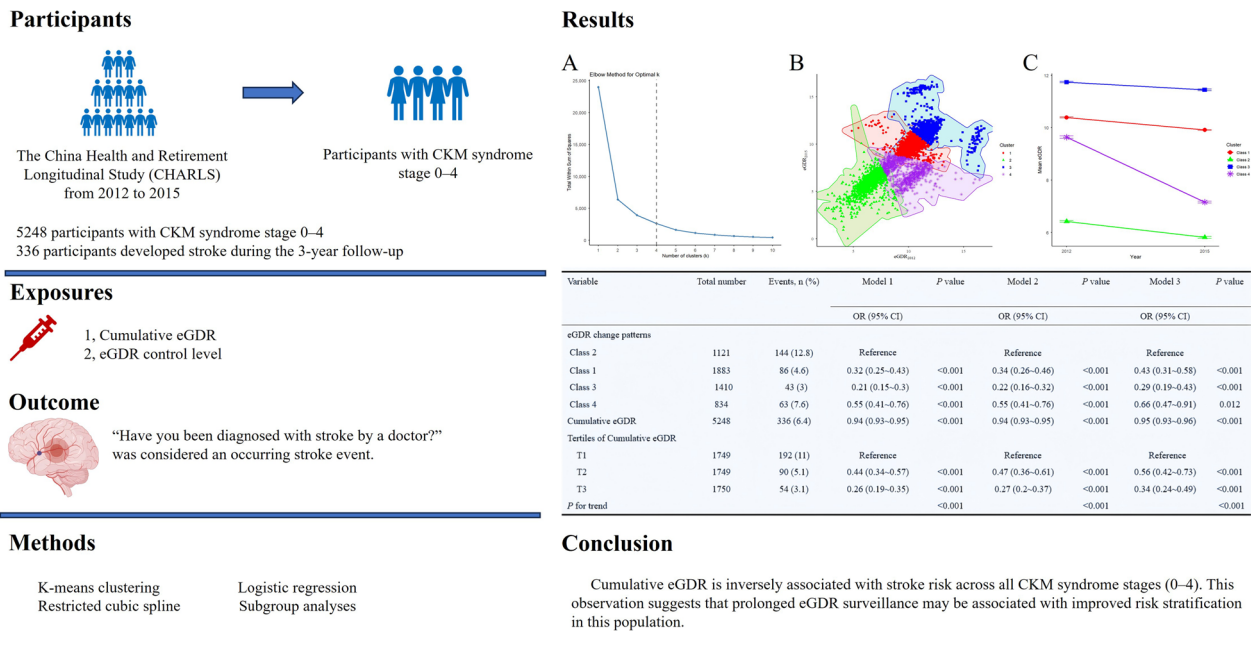


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Conclusion Cumulative eGDR is inversely associated with stroke risk across all CKM syndrome stages (0–4). This observation suggests that prolonged eGDR surveillance may be associated with improved risk stratification in this population.

Keywords Cumulative estimated glucose disposal rate, Stroke, Cardiovascular–kidney–metabolic syndrome

Graphical Abstract



Introduction

Stroke poses a critical public health threat in China and is characterised by substantial and growing burdens of morbidity and disability [1, 2]. Cardiovascular–kidney–metabolic (CKM) syndrome—a pathophysiological nexus interconnecting obesity, diabetes, and chronic kidney disease—increasingly frames our understanding of this challenge by amplifying vulnerability to cardio-cerebrovascular events [3]. Susceptibility linked to CKM syndrome is not static but follows a graded progression; individuals in stages 3–4 face a disproportionately greater stroke risk than those in stages 0–2 [4]. This graded risk pattern underscores the necessity of identifying modifiable risk factors, which can inform early intervention strategies for stroke prevention in the CKM population.

The estimated glucose disposal rate (eGDR), a proxy measure of peripheral insulin sensitivity, is computed as follows: $eGDR = 21.158 - (0.090 \times WC [cm]) - (3.407 \times HTN [presence = 1]) - (0.551 \times HbA1c [\%])$, with higher values signifying increased insulin sensitivity [5]. Low eGDR not only corresponds to severe insulin resistance but is also related to stroke occurrence through multiple biological mechanisms [6, 7]. At the vascular level, insulin resistance compromises endothelial function by reducing nitric oxide synthesis and increasing endothelin-1 production, leading to impaired

vasodilation, increased vasoconstriction, and altered vascular reactivity. This endothelial dysfunction promotes atherosclerotic plaque formation and destabilisation, which are critical events in stroke pathogenesis [8–10]. Systemically, it perpetuates a chronic inflammatory state characterised by increased cytokine levels—including high-sensitivity C-reactive protein (hs-CRP), interleukin-6 (IL-6), and tumour necrosis factor- α (TNF- α)—which together impair endothelial integrity, accelerate atherogenesis, and foster a prothrombotic milieu [10–12]. Furthermore, insulin resistance exacerbates oxidative stress through dual pathways: increasing reactive oxygen species (ROS) generation and decreasing antioxidant defences, culminating in vascular cellular damage and dysfunction [13]. These pathways are further amplified by associated metabolic disturbances—including dyslipidaemia, sympathetic nervous system activation, and hypercoagulability mediated by elevated plasminogen activator inhibitor-1—collectively increasing susceptibility to cerebrovascular events [14–16].

Substantial evidence indicates that reduced eGDR contributes to cerebrocardiovascular pathogenesis. Dong et al. [17] reported a 42% increase in cardiovascular disease risk (HR = 1.42, 95% CI 1.36–1.48) per 1-unit eGDR decrease in patients with CKM syndrome stages 0–3. Similarly, Zabala et al. [18] reported a 78% higher stroke

incidence in the lowest vs. highest eGDR quartile (HR 1.78, 95% CI 1.54–2.06) among 32,087 T2DM patients. Liang et al. [19] revealed a U-shaped relationship: cardiovascular risk was increased in both the lowest eGDR stratum (HR 2.11, 95% CI 1.76–2.53) and the highest stratum (HR 1.48, 1.12–1.96), indicating adverse effects at eGDR extremes. This relationship was further supported by a systematic review and meta-analysis by Shojaei et al. [20], encompassing more than 1.2 million participants, in which lower eGDR was consistently linked to major adverse cardio-cerebrovascular events (RR = 1.84; 95% CI: 1.67–2.03).

However, given that eGDR is a dynamic parameter, a single assessment may not fully capture its cumulative burden. Although cumulative exposure to cardiovascular risk factors—not baseline levels—is strongly associated with long-term clinical outcomes, the influence of cumulative eGDR on stroke in the context of CKM syndrome has not been specifically assessed [4, 6]. This study examined how longitudinal eGDR variations are related to stroke incidence across all stages of CKM syndrome (0–4) in the CHARLS cohort, addressing a critical gap in evidence.

Methods

Study population

CHARLS (China Health and Retirement Longitudinal Study), a national population-based cohort, enrolled Chinese participants aged ≥ 45 years. This study was carried out jointly by Peking University's National School of Development and the Chinese Academy of Social Sciences. It utilised a multistage, stratified sampling approach, with four waves of data collection conducted in 2012 (baseline), 2013, 2015, and 2018. All surveys followed standardised protocols to maintain data quality and comparability. Further methodological details are available in prior publications [21, 22].

Participants were included if they met the following criteria: (1) complete data on CKM syndrome staging (stages 0–4); (2) no stroke history and available stroke status data from waves 1 to 3 (2012–2015); (3) accessible stroke outcome data at wave 4 (2018); and (4) complete data for estimated glucose disposal rate (eGDR) calculation (without outliers) at waves 1 or 3. Individuals not meeting all these criteria were excluded from the analysis.

Ethical approval for the CHARLS study was obtained from the IRB of Peking University (IRB00001052-11015). All participants provided written informed consent prior to enrolment.

Assessment of eGDR, cumulative eGDR, and eGDR change patterns

Waist circumference (WC) was measured at the umbilical level in standing participants during normal tidal breathing [23]. Hypertension (HTN) was defined by: (1) prior physician diagnosis, (2) current antihypertensive treatment, or (3) blood pressure $\geq 140/90$ mmHg. Glycated haemoglobin (HbA1c) measurements from whole blood samples underwent NHANES-standardised assay calibration [24]. eGDR was derived using the validated equation: $21.158 - [0.090 \times \text{WC (cm)}] - [3.407 \times \text{HTN (presence = 1)}] - [0.551 \times \text{HbA1c (\%)}]$. Here, WC (cm), HTN (1 = present; 0 = absent), and HbA1c (%) operationalised these variables [5, 25]. Cumulative eGDR exposure was computed as follows: $(\text{eGDR}_{2012} + \text{eGDR}_{2015})/2 \times (\text{time}_{2015} - \text{time}_{2012})$ [26]. Participants were subsequently classified into four distinct eGDR change patterns through k-means clustering analysis using eGDR values from both 2012 and 2015 as input features: Class 1 (moderate–high stable), Class 2 (persistent low), Class 3 (stable high), and Class 4 (rapid decrease).

CKM syndrome staging classification (0–4)

CKM syndrome staging at baseline (2012, Wave 1) was determined following the American Heart Association framework, which categorises the condition into five progressive stages (0–4) [3]. Stage 0 characterises individuals without metabolic or cardiovascular risk factors; Stage 1 manifests overweight/obesity or prediabetes;

Stage 2 is defined by established conditions, including type 2 diabetes (T2DM), hypertension (HTN), hypertriglyceridaemia, and chronic kidney disease (CKD); Stage 3 involves subclinical cardiovascular impairment, such as asymptomatic atherosclerosis or left ventricular dysfunction; and Stage 4 represents clinical cardiovascular disease—such as coronary artery disease (CAD), heart failure (HF), stroke, peripheral artery disease (PAD), and atrial fibrillation (AF)—in patients with established CKM pathophysiology.

Outcome ascertainment

The incidence of stroke during follow-up served as the primary outcome. In Wave 4, stroke events were ascertained when participants reported affirmative responses to the physician-diagnosed question: “Have you been diagnosed with stroke by a doctor?” This case identification approach has been validated in prior investigations [27, 28].

Data collection

At baseline (Wave 1, 2011–2012), trained personnel administered standardised questionnaires to obtain sociodemographic and health-related information. Collected variables included age, sex, marital status, and

education level—with education categorised as no formal education, $\leq$ middle school, high school or vocational training, or $\geq$ college. Smoking status and alcohol use status were dichotomised as never/ever exposure. Physician-diagnosed hypertension, diabetes mellitus, and dyslipidaemia were additionally ascertained through self-reports. Physical examinations revealed the following parameters: body mass index (BMI), waist circumference (WC), systolic blood pressure (SBP), and diastolic blood pressure (DBP). Additional laboratory assays quantified triglyceride (TG) levels, blood glucose levels, HbA1c levels, HDL cholesterol levels, LDL cholesterol levels, uric acid (UA) levels, eGFRs, and total cholesterol (TC) levels.

Statistical analysis

Continuous variables are presented as mean $\pm$ SD when normally distributed or as median (IQR) otherwise. Categorical data are reported as counts and percentages. Baseline comparisons were performed using χ^2 /Fisher exact tests (expected frequencies < 5) for categorical variables, t-test for normally distributed continuous variables, and Mann–Whitney U tests for skewed continuous variables.

To characterise longitudinal eGDR patterns from 2012 to 2015, we applied the k-means clustering algorithm using eGDR measurements from both time points (2012 and 2015) as input variables for each participant. This bivariate approach enabled the algorithm to capture heterogeneity in both baseline insulin sensitivity status and temporal evolution. The k-means method, an unsupervised machine learning technique, groups observations by minimising within-cluster variance. The optimal number of clusters was determined using the elbow method, which evaluates the reduction in within-cluster sum of squares as the number of clusters increases. A clear inflection point emerged at $K=4$, which suggests that four clusters provided an optimal balance between model fit and complexity (Fig. 2A). On the basis of this solution, the participants were categorised into four distinct trajectory groups on the basis of their proximity to the final cluster centroids (Fig. 2B).

Cumulative eGDR exposure was analysed as both a continuous measure and in tertiles to evaluate its dose-response relationship with stroke incidence. Given the low incidence of stroke (6.40%) in our cohort, multivariable logistic regression was used to estimate odds ratios (ORs), which approximate hazard ratios (HRs) under the rare disease assumption [29]. To address potential limitations from follow-up duration, we conducted Cox proportional hazards models as sensitivity analyses. Results are reported as odds ratios (ORs) and 95% confidence intervals (CIs) across sequential models: Model 1 (unadjusted crude analysis); Model 2 (adjusted for demographic covariates: age, sex, and marital status); and

Model 3 (further adjusted for behavioural and clinical confounders, including education, smoking status, alcohol use, body mass index, dyslipidaemia, diabetes, and estimated glomerular filtration rate). To facilitate clinical interpretation, the effect size for continuous cumulative eGDR is reported per 1-unit increase. Potential nonlinear associations were examined using restricted cubic splines.

Effect modification was evaluated via stratified analyses across these subgroups: age (< 60 vs. ≥ 60 years), sex, educational attainment (below high school vs. high school or higher), behavioural factors (never vs. ever tobacco and alcohol exposure), and CKM stage (0–2 vs. 3–4), and the presence of diabetes, hypertension, or dyslipidaemia. All stratified models retained the covariates specified in Model 3, and interaction terms were tested using likelihood ratio tests. To assess the robustness of our findings, we conducted two sensitivity analyses: (1) reclassifying cumulative eGDR into tertiles and testing for linear trends, and (2) multiple imputation via chained equations (MICE; five datasets) was applied to manage missing covariate data, followed by comparison of pooled estimates with complete-case results. Analyses were performed using R 4.2.2 and Free Statistics software v2.0, with statistical significance set at two-tailed $P < 0.05$.

Results

Baseline characteristics by eGDR change classes

Figure 1 depicts the flow diagram of participant selection. The CHARLS baseline cohort (2011–2012) initially comprised 17,708 eligible participants. Participants with incomplete CKM syndrome staging data ($n=8307$) were excluded, leaving 9401 individuals with documented CKM stages 0–4. Among these, sequential exclusions were applied for the following: history of stroke or missing stroke status during the baseline and follow-up period from waves 1 through 3 ($n=1828$); absence of stroke outcome data at wave 4 ($n=710$); and incomplete eGDR measurements or outlier values at baseline or wave 3 ($n=1611$). After these exclusion criteria were applied, the final analytical cohort comprised 5248 participants categorised into four classes on the basis of eGDR change patterns (Class 1, $n=1883$; Class 2, $n=1121$; Class 3, $n=1410$; and Class 4, $n=834$).

Table 1 presents the baseline characteristics of participants with CKM syndrome (stages 0–4), stratified by four eGDR change pattern groups (classes 1–4). Specifically, Class 1 included 1883 participants, with eGDR values ranging from 10.39 ± 0.75 in 2012 to 9.91 ± 0.63 in 2015, which is consistent with a stable moderate-risk group; Class 2 comprised 1121 participants, whose eGDR values ranged from 6.42 ± 1.00 in 2012 to 5.81 ± 1.15 in 2015, corresponding to a persistent high-risk group; Class 3 had 1410 participants, with eGDR values ranging

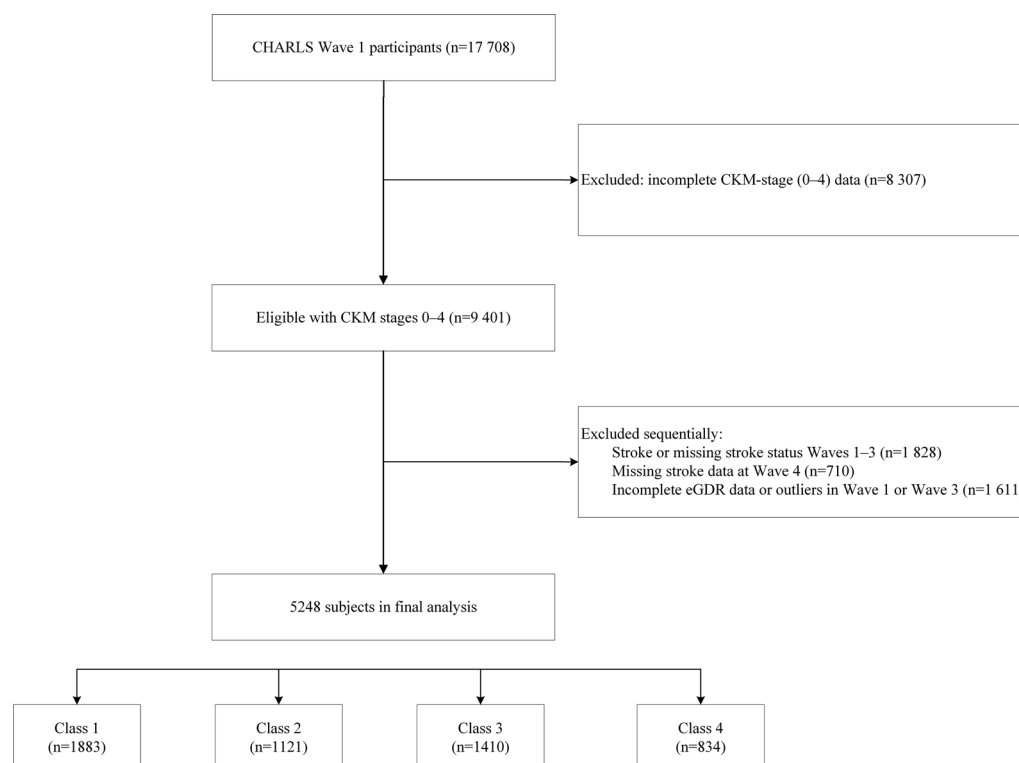


Fig. 1 Flowchart of the study population

from 11.74 ± 1.04 in 2012 to 11.45 ± 1.18 in 2015, which was categorised as a stable low-risk group; and Class 4 included 834 participants, whose eGDR values decreased from 9.64 ± 1.35 in 2012 to 7.16 ± 1.03 in 2015, defined as a rapid decrease group.

Males accounted for 43.73% of participants. Mean eGDR values for the study population were 9.79 ± 2.14 in 2012 and 9.01 ± 2.36 in 2015, with a cumulative eGDR of 37.54 ± 8.46 . Significant intergroup differences emerged in demographics (age, sex, and BMI), sociobehavioural factors (education and smoking/alcohol status), and clinical stage (CKM severity). The prevalence of comorbidities (hypertension, diabetes, and dyslipidaemia) and metabolic parameters (e.g., blood glucose level, glycated haemoglobin level, lipid profile, blood pressure, and waist circumference) also differed significantly across the groups.

Specifically, Class 2 was the group with the most severe metabolic dysregulation: compared with the other classes, it had the highest BMI (26.40 ± 4.17 kg/m²), the highest prevalence of comorbidities (96.34% for hypertension, 15.19% for diabetes, 24.29% for dyslipidaemia), and the most prominent abnormalities in metabolic parameters (SBP 146.61 ± 22.77 mmHg, DBP 83.69 ± 12.84 mmHg, WC 93.46 ± 8.57 cm, glucose 120.58 ± 51.62 mg/dL, HbA1c $5.52 \pm 1.09\%$, TG 164.53 ± 140.56 mg/dL). In contrast, Class 3 had the most favourable metabolic profile, with the lowest waist

circumference (73.85 ± 11.40 cm) and the highest HDL cholesterol concentration (57.12 ± 15.82 mg/dL) among all classes (Table 1).

Key findings of K-means clustering for eGDR changes and the identification of the optimal number of clusters are presented in Fig. 2. Clustering was performed using eGDR values from both 2012 and 2015 as input features. The elbow method plot is shown in Fig. 2A, with within-cluster sum of squares (WCSS) on the y-axis versus cluster number (k) on the x-axis; a clear elbow was observed at $k=4$, at which point the reduction in within-cluster sum of squares levelled off. This confirmed that 4 clusters were the optimal choice to balance the heterogeneity of eGDR changes and model simplicity. The eGDR values of the 4 identified clusters (class 1 to class 4) at two time points (2012 and 2015) are shown in Fig. 2B. Classes 1, 2, and 3 maintained relatively stable eGDR levels between these two years, whereas class 4 exhibited a distinct downwards trend in eGDR. These results provide a basis for subsequent subgroup analyses related to eGDR changes and stroke risk. The distribution of eGDR values across the four clusters at both time points (2012 and 2015) is shown in Fig. 2C. These patterns, identified using two-time point data, provide a basis for subsequent subgroup analyses of eGDR changes and stroke risk.

Table 1 Baseline characteristics according to eGDR change patterns

Characteristics	Overall (N = 5248)	Class 1 (N = 1883)	Class 2 (N = 1121)	Class 3 (N = 1410)	Class 4 (N = 834)	P-value
Age, mean (SD), years	58.29 (8.86)	57.21 (8.69)	59.83 (8.58)	57.92 (9.25)	59.31 (8.58)	< 0.001
Gender, n (%)						< 0.001
Female	2953 (56.27%)	1068 (56.72%)	698 (62.27%)	746 (52.91%)	441 (52.88%)	
Male	2295 (43.73%)	815 (43.28%)	423 (37.73%)	664 (47.09%)	393 (47.12%)	
BMI, mean (SD), kg/m ²	23.68 (3.98)	24.01 (3.10)	26.40 (4.17)	20.78 (2.81)	24.22 (4.05)	< 0.001
Educational level, n (%)						0.004
No formal education	2493 (47.52%)	832 (44.21%)	535 (47.73%)	696 (49.40%)	430 (51.56%)	
Middle school or below	1200 (22.87%)	428 (22.74%)	252 (22.48%)	326 (23.14%)	194 (23.26%)	
High school or vocational school	1066 (20.32%)	420 (22.32%)	231 (20.61%)	269 (19.09%)	146 (17.51%)	
College or above	487 (9.28%)	202 (10.73%)	103 (9.19%)	118 (8.37%)	64 (7.67%)	
Smoking status, n (%)						< 0.001
Never	3315 (63.17%)	1221 (64.84%)	766 (68.33%)	835 (59.22%)	493 (59.11%)	
Ever	1933 (36.83%)	662 (35.16%)	355 (31.67%)	575 (40.78%)	341 (40.89%)	
Drinking status, n (%)						0.037
Never	3276 (62.47%)	1186 (63.02%)	733 (65.45%)	860 (61.04%)	497 (59.66%)	
Ever	1968 (37.53%)	696 (36.98%)	387 (34.55%)	549 (38.96%)	336 (40.34%)	
Marital status, n (%)						0.092
Single	517 (9.85%)	172 (9.13%)	113 (10.08%)	131 (9.29%)	101 (12.11%)	
Married	4731 (90.15%)	1711 (90.87%)	1008 (89.92%)	1279 (90.71%)	733 (87.89%)	
SBP, mean (SD), mm Hg	131.47 (21.93)	126.12 (18.03)	146.61 (22.77)	121.79 (17.18)	139.74 (22.39)	< 0.001
DBP, mean (SD), mm Hg	76.36 (12.63)	74.37 (11.22)	83.69 (12.84)	71.22 (10.97)	79.77 (12.53)	< 0.001
WC, mean (SD), cm	84.61 (12.29)	87.09 (6.85)	93.46 (8.57)	73.85 (11.40)	85.32 (14.55)	< 0.001
Glucose, mean (SD), mg/dl	109.33 (35.23)	105.89 (26.11)	120.58 (51.62)	100.74 (17.47)	116.48 (42.51)	< 0.001
TC, mean (SD), mg/dL	193.36 (38.24)	193.47 (36.66)	200.13 (40.89)	186.48 (35.89)	195.60 (40.01)	< 0.001
TG, mean (SD), mg/dL	133.54 (107.22)	134.25 (100.14)	164.53 (140.56)	101.42 (66.80)	144.57 (111.70)	< 0.001
HDL, mean (SD), mg/dL	51.05 (15.16)	49.71 (13.84)	46.42 (13.75)	57.12 (15.82)	50.04 (15.52)	< 0.001
LDL, mean (SD), mg/dL	116.28 (34.68)	117.47 (33.95)	121.00 (37.87)	110.30 (30.94)	117.36 (36.49)	< 0.001
HbA1c, mean (SD), %	5.26 (0.77)	5.21 (0.59)	5.52 (1.09)	5.03 (0.40)	5.39 (0.96)	< 0.001
Uric acid, mean (SD), mg/dl	4.34 (1.20)	4.28 (1.16)	4.61 (1.26)	4.15 (1.13)	4.42 (1.26)	< 0.001
eGFR, mL/min/1.73 m ²	93.52 (14.09)	94.75 (13.56)	90.21 (14.60)	95.10 (13.50)	92.54 (14.75)	< 0.001
Cumulative eGDR	37.54 (8.46)	40.56 (2.24)	24.47 (3.90)	46.26 (2.98)	33.55 (2.97)	< 0.001
eGDR ₂₀₁₂ , mean (SD)	9.79 (2.14)	10.39 (0.75)	6.42 (1.00)	11.74 (1.04)	9.64 (1.35)	< 0.001
eGDR ₂₀₁₅ , mean (SD)	9.01 (2.36)	9.91 (0.63)	5.81 (1.15)	11.45 (1.18)	7.16 (1.03)	< 0.001
Hypertension, n (%)	1326 (25.27%)	31 (1.65%)	1080 (96.34%)	1 (0.07%)	214 (25.66%)	< 0.001
Diabetes, n (%)	304 (5.80%)	51 (2.71%)	170 (15.19%)	20 (1.42%)	63 (7.57%)	< 0.001
Dyslipidemia, n (%)	516 (9.96%)	128 (6.88%)	267 (24.29%)	42 (3.01%)	79 (9.58%)	< 0.001
Stroke, n (%)	336 (6.40%)	86 (4.57%)	144 (12.85%)	43 (3.05%)	63 (7.55%)	< 0.001
CKM syndrome stages, n (%)						< 0.001
0	196 (3.73%)	28 (1.49%)	0 (0.00%)	162 (11.49%)	6 (0.72%)	
1	576 (10.98%)	322 (17.10%)	0 (0.00%)	219 (15.53%)	35 (4.20%)	
2	1352 (25.76%)	526 (27.93%)	326 (29.08%)	268 (19.01%)	232 (27.82%)	
3	2462 (46.91%)	837 (44.45%)	535 (47.73%)	657 (46.60%)	433 (51.92%)	
4	662 (12.61%)	170 (9.03%)	260 (23.19%)	104 (7.38%)	128 (15.35%)	

Continuous variables are presented as means (SD) and categorical variables are presented as n (%). SD, standard deviations; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; WC, waist circumference; TC, total cholesterol; TG, triglycerides; HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol; HbA1c, glycosylated hemoglobin type A1c; eGFR, estimated glomerular filtration rate; eGDR, estimated glucose disposal rate; CKM, cardiovascular-kidney-metabolic

Associations of eGDR change patterns and cumulative exposure with incident stroke across CKM syndrome stages 0–4

Table 2 shows the associations between eGDR change patterns, cumulative eGDR exposure, and incident stroke

in participants with CKM syndrome (stages 0–4) on the basis of multivariable logistic regression models. Additionally, Cox proportional hazards regression analysis is provided in Supplementary Table S1. In Model 3 (fully adjusted; Class 2 reference), Class 1 had a significantly

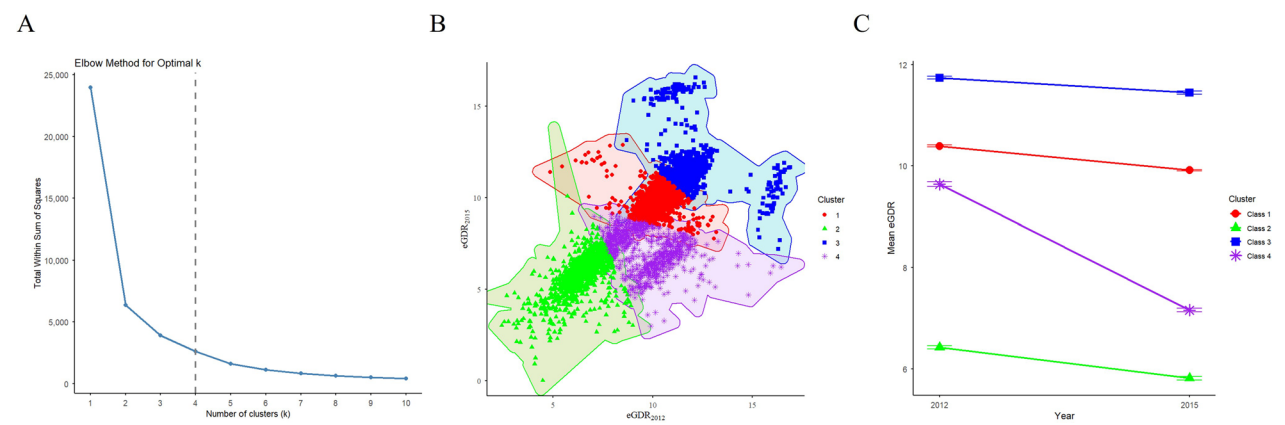


Fig. 2 Clustering of eGDR Changes from Wave 1 to Wave 3. K-means clustering was performed using eGDR values from both 2012 and 2015 as input features. **A** Elbow method plot showing the optimal number of clusters (k=4) based on within-cluster sum of squares; **B** Mean eGDR trajectories of the four classes from 2012 to 2015; **C** Distribution of eGDR values at 2012 and 2015 timepoints across the four clusters. Class 1 (Red)—Moderate-High Stable Group; Class 2 (Green)—Persistent Low Group; Class 3 (Blue)—Stable High Group; Class 4 (Purple)—Rapid Decrease Group

Table 2 Logistic regression for associations between eGDR change patterns, cumulative eGDR, and stroke in subjects with CKM syndrome stages 0–4

Variable	Total number	Events, n (%)	Model 1	P value	Model 2	P value	Model 3	P value
			OR (95% CI)		OR (95% CI)		OR (95% CI)	
eGDR change patterns								
Class 2	1121	144 (12.8)	Reference		Reference		Reference	
Class 1	1883	86 (4.6)	0.32 (0.25–0.43)	< 0.001	0.34 (0.26–0.46)	< 0.001	0.43 (0.31–0.58)	< 0.001
Class 3	1410	43 (3)	0.21 (0.15–0.3)	< 0.001	0.22 (0.16–0.32)	< 0.001	0.29 (0.19–0.43)	< 0.001
Class 4	834	63 (7.6)	0.55 (0.41–0.76)	< 0.001	0.55 (0.41–0.76)	< 0.001	0.66 (0.47–0.91)	0.012
Cumulative eGDR	5248	336 (6.4)	0.94 (0.93–0.95)	< 0.001	0.94 (0.93–0.95)	< 0.001	0.95 (0.93–0.96)	< 0.001
Tertiles of cumulative eGDR								
T1	1749	192 (11)	Reference		Reference		Reference	
T2	1749	90 (5.1)	0.44 (0.34–0.57)	< 0.001	0.47 (0.36–0.61)	< 0.001	0.56 (0.42–0.73)	< 0.001
T3	1750	54 (3.1)	0.26 (0.19–0.35)	< 0.001	0.27 (0.2–0.37)	< 0.001	0.34 (0.24–0.49)	< 0.001
P for trend				< 0.001		< 0.001		< 0.001

Model 1: No covariates were adjusted

Model 2: Adjusted for age, gender and marital status

Model 3: Adjusted for age, gender, marital status, BMI, educational level, smoke status, drink status, eGFR, dyslipidemia, diabetes

CI confidence interval; OR odds ratio; T tertiles; eGDR estimated glucose disposal rate; BMI body mass index; eGFR estimated glomerular filtration rate; CKM cardiovascular-kidney-metabolic

lower stroke risk (OR 0.43, 95% CI: 0.31–0.58; $P < 0.001$), with similar reductions in Class 3 (0.29, 0.19–0.43; $P < 0.001$) and Class 4 (0.66, 0.47–0.91; $P = 0.012$). Per-unit increase in cumulative eGDR (continuous) was inversely associated with stroke risk (OR 0.95, 0.93–0.96; $P < 0.001$). Cumulative eGDR tertiles revealed a dose-dependent stroke risk reduction: compared with T1 (reference), T2 had an OR of 0.56 (0.42–0.73; $P < 0.001$), and T3 had an OR of 0.34 (0.24–0.49; $P < 0.001$), with a trend of $P < 0.001$. Restricted cubic spline regression revealed a linear inverse relationship between cumulative eGDR and stroke risk (Fig. 3).

Subgroup analyses of eGDR patterns and stroke risk in patients with CKM syndrome

Table 3 presents subgroup analyses of the association between eGDR change patterns (classes 1–4) and stroke incidence, stratified by demographic factors (age, sex, and BMI), behavioural factors (smoking/drinking status), and clinical factors (dyslipidaemia, diabetes, and CKM stage). All models were adjusted for the full set of covariates except the stratification variable itself, with Class 2 serving as the reference. A statistically significant interaction was identified exclusively for diabetes status, with an interaction P value of 0.002. Among participants without diabetes, both the moderate–high stable (class 1: OR 0.45; 95% CI: 0.32–0.61) and stable high (class 3: OR 0.29; 95% CI: 0.19–0.45) patterns were associated with a significantly reduced stroke risk. In contrast, effect estimates

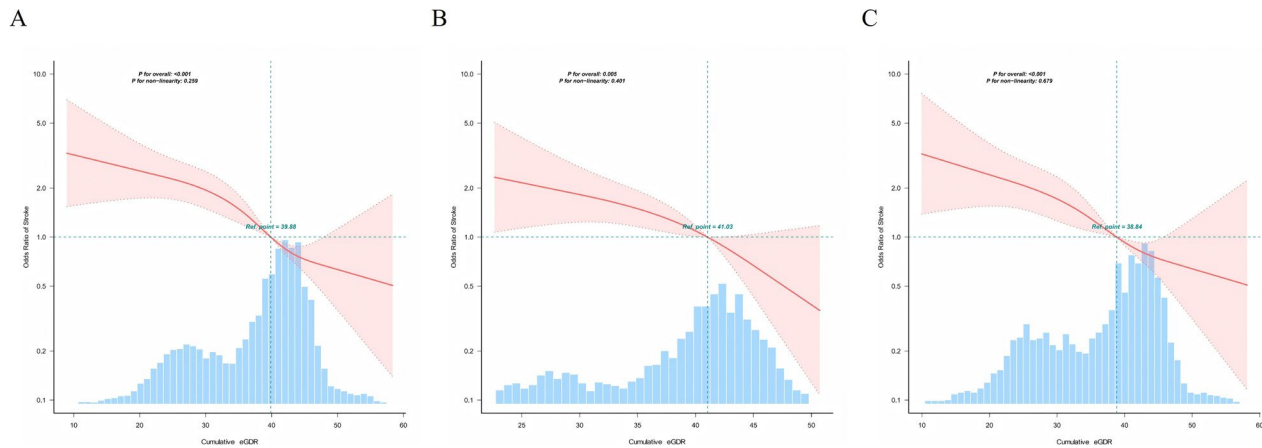


Fig. 3 The RCS analysis of the relationship between cumulative eGDR and stroke incidence in participants with CKM syndrome (stages 0–4). Adjustment variables included for age, gender, marital status, body mass index (BMI), educational attainment, smoking status, alcohol consumption status, estimated glomerular filtration rate (eGFR), dyslipidemia, and diabetes. **A** Relationship between cumulative eGDR and stroke incidence in participants with CKM syndrome (Stages 0–4); **B** Relationship between cumulative eGDR and stroke incidence in participants with non-advanced CKM syndrome (Stages 0–2); **C** Relationship between cumulative eGDR and stroke incidence in participants with advanced CKM syndrome (stages 3–4). BMI, body mass index; eGFR, estimated glomerular filtration rate; CKM, cardiovascular-kidney-metabolic; OR, odds ratio

Table 3 Subgroup analysis of eGDR change patterns and stroke incidence in subjects with CKM syndrome (stages 0–4)

Subgroup	Class2	Class 1	Class 3	Class 4	P for trend	P for interaction
Age, years						0.512
< 60	Reference	0.35 (0.23–0.54)	0.21 (0.11–0.38)	0.53 (0.33–0.87)	0.004	
≥ 60	Reference	0.49 (0.32–0.75)	0.37 (0.22–0.64)	0.78 (0.5–1.21)	0.158	
Gender						0.261
Female	Reference	0.33 (0.22–0.5)	0.3 (0.18–0.51)	0.6 (0.39–0.93)	0.006	
Male	Reference	0.59 (0.36–0.96)	0.29 (0.15–0.56)	0.8 (0.48–1.33)	0.388	
BMI, kg/m ²						0.543
< 30	Reference	0.43 (0.31–0.58)	0.29 (0.2–0.42)	0.68 (0.49–0.96)	0.001	
≥ 30	Reference	0.49 (0.1–2.42)	0 (0–Inf)	0.27 (0.07–1)	0.03	
Smoke						0.823
Never	Reference	0.4 (0.27–0.58)	0.3 (0.18–0.51)	0.63 (0.41–0.96)	0.008	
Ever	Reference	0.47 (0.28–0.78)	0.27 (0.14–0.52)	0.71 (0.42–1.19)	0.167	
Drink						0.909
Never	Reference	0.43 (0.29–0.63)	0.28 (0.17–0.48)	0.6 (0.38–0.92)	0.004	
Ever	Reference	0.41 (0.25–0.68)	0.29 (0.15–0.55)	0.75 (0.45–1.26)	0.265	
Dyslipemia						0.137
No	Reference	0.48 (0.34–0.68)	0.32 (0.21–0.5)	0.76 (0.53–1.09)	0.083	
Yes	Reference	0.24 (0.1–0.57)	0.11 (0.01–0.85)	0.4 (0.17–0.96)	0.005	
Diabetes						0.002
No	Reference	0.45 (0.32–0.61)	0.29 (0.19–0.45)	0.76 (0.54–1.07)	0.056	
Yes	Reference	0.27 (0.07–1.08)	0.53 (0.09–2.96)	0 (0–Inf)	0.005	
CKM						0.562
0–2	Reference	0.49 (0.28–0.87)	0.39 (0.19–0.82)	0.99 (0.54–1.87)	0.961	
3–4	Reference	0.43 (0.3–0.63)	0.28 (0.17–0.46)	0.57 (0.39–0.85)	0.002	

Adjusted for all covariates except for this subgroup of variables

CI confidence interval; OR odds ratio; BMI body mass index; CKM cardiovascular-kidney-metabolic

among diabetic patients were unstable: the associations for Class 1 (OR 0.27, 95% CI: 0.07–1.08) and Class 3 (OR 0.53, 95% CI: 0.09–2.96) were nonsignificant, and a valid odds ratio for Class 4 could not be estimated because of an insufficient number of stroke events in this subgroup.

No other subgroup variables significantly modified the effect ($P > 0.05$ for all interactions). Cox regression analyses are presented in Supplementary Table S2. Sensitivity analyses consistently revealed an inverse association between cumulative eGDR and stroke risk (Figs. S1, S2,

S3; Tables S1, S3 and S4). These findings align with methodological guidance that OR approximates HR when event incidence is < 10% [30].

Discussion

This study investigated the associations between stroke risk and eGDR change patterns and cumulative eGDR across cardiovascular–kidney–metabolic (CKM) syndrome stages 0–4. Using k-means clustering, we identified four distinct eGDR patterns over the 3-year observation period. Compared with the reference group with persistently low eGDR levels (Class 2), which exhibited the highest stroke incidence (12.85%), all other groups showed substantially reduced stroke risk after multivariable adjustment. Additionally, cumulative eGDR was independently associated with stroke occurrence, with a dose-dependent relationship observed across tertiles. Participants with higher cumulative eGDR values experienced progressively lower stroke risk. These findings suggest that long-term maintenance of elevated eGDR levels is associated with reduced stroke risk in this population.

The current results align with those of previous prospective cohort investigations demonstrating an inverse relationship between the eGDR and incident cardio-cerebrovascular outcomes [31, 32]. For instance, an analysis of the CHARLS by Jiang et al. revealed that a per-SD eGDR increase resulted in a 25.4% lower stroke risk in dysglycaemic adults ≥ 45 years [33]. Similarly, Tao et al. [34] reported a 39.5% reduction in stroke incidence per 1-unit eGDR increase in a general Chinese population. This pattern is further supported by Zhang et al.'s analysis of the UK Biobank cohort, which revealed progressively lower cardiovascular disease occurrence across ascending eGDR quartiles among participants in CKM syndrome stages 0–3 [35]. The consistency of this relationship across populations with varying metabolic characteristics strengthens the evidence for an interconnection between insulin sensitivity and cerebrovascular health.

Our investigation expands upon existing research through its specific focus on CKM syndrome stages 0–4 and longitudinal eGDR assessment. While previous reports primarily utilised single eGDR measurements [33, 36], our analysis of eGDR patterns over time provides a new understanding of the dynamic interconnection between insulin sensitivity and stroke occurrence in this population with multisystem involvement. This methodological approach, involving k-means clustering of serial eGDR measurements, builds upon the analytical framework demonstrated in prior longitudinal studies of metabolic indices [4]. Furthermore, our inclusion of individuals across the complete CKM spectrum, including stage 4, represents an important extension beyond earlier

work limited to initial disease phases [37], enabling a more thorough characterisation of the eGDR-stroke interconnection throughout CKM progression.

This inverse eGDR-stroke risk association can be attributed to multiple interconnected biological pathways, reflecting the complex relationship between insulin resistance (IR) and cerebrovascular pathophysiology. IR contributes to endothelial dysfunction by impairing nitric oxide (NO) synthesis while promoting endothelin-1 release, which collectively disrupts vascular homeostasis and facilitates atherosclerotic plaque instability [38, 39]. The link between eGDR and stroke further underscores the importance of NO in preserving endothelial integrity and modulating inflammatory responses, as demonstrated in studies by Huang et al. [40]. Their work confirmed that reduced eGDR is associated with increased inflammatory markers, highlighting the involvement of immune pathways in stroke pathogenesis [41]. Key proinflammatory mediators—hs-CRP, IL-6, and TNF- α —increase vascular risk by exacerbating endothelial injury, promoting thrombus formation, and accelerating atherosclerosis [20].

This study has several methodological strengths that distinguish it from prior investigations. First, unlike previous research reliant on single measurements, we identified clinically meaningful subphenotypes by applying k-means clustering to baseline eGDR, revealing heterogeneous risk profiles that were not apparent when the population was treated as homogeneous. Second, the combination of this baseline stratification with cumulative eGDR exposure represents a more nuanced approach to assessing insulin resistance burden, capturing both cross-sectional heterogeneity and temporal exposure. Third, in contrast to prior studies that were largely confined to diabetic cohorts or excluded patients with cardiovascular disease (CKM stage 4), our investigation captured the complete CKM continuum (stages 0–4), thereby strengthening the applicability of the results to unselected clinical populations. Fourth, we minimised residual confounding through sequential adjustments for sociodemographic, lifestyle, anthropometric, and cardiometabolic factors in multivariable models. Finally, sensitivity analyses validated the stability of the observed associations, reinforcing the robustness of our findings. These results support the hypothesis that eGDR monitoring may help identify individuals at high risk of stroke.

Several limitations warrant consideration. First, although the low event rate supports the use of logistic regression, our primary analysis did not account for the exact timing of events or potential censoring. However, since we conducted only one follow-up assessment from 2015 to 2018, there are no issues related to censoring in our analysis. Future studies with extended follow-up should employ time-to-event models to fully capture

temporal dynamics [42]. Second, the final analytical cohort exclusively comprised adults aged above 45 years, which may limit generalisability to younger or ethnically diverse populations. In addition, the observational nature of this study fundamentally precludes causal inference. Despite extensive multivariable adjustment, residual confounding from unmeasured factors cannot be excluded. Moreover, reverse causality remains a possibility: sub-clinical disease may lead to changes in body weight or physical activity, which in turn influence eGDR values. These bidirectional relationships cannot be definitively disentangled in our observational framework. Fourth, stroke outcomes were ascertained through self-reports rather than standardised clinical adjudication, introducing the possibility of diagnostic misclassification. Finally, the 6-year observation window might not be adequate for evaluating the long-term cerebrovascular consequences associated with insulin resistance. Future research involving extended follow-up durations, objectively verified endpoints, and more heterogeneous populations is warranted to validate our observations.

Conclusion

In individuals with CKM syndrome stages 0–4, a significant inverse association was observed between cumulative eGDR and stroke risk. These findings suggest that identifying distinct eGDR trajectory patterns may help stratify stroke risk in this population.

Abbreviations

BMI	Body mass index
CKM	Cardiovascular–kidney–metabolic
CI	Confidence interval
CRP	C-reactive protein
DBP	Diastolic blood pressure
DM	Diabetes mellitus
eGDR	Estimated glucose disposal rate
eGFR	Estimated glomerular filtration rate
WC	Waist circumference
HbA1c	Glycosylated hemoglobin type A1c
HDL	High-density lipoprotein cholesterol
HR	Hazard ratio
IQR	Interquartile range
LDL	Low-density lipoprotein cholesterol
IL-6	Interleukin-6
OR	Odds ratio
RCS	Restricted cubic spline
SBP	Systolic blood pressure
TC	Total cholesterol
TG	Triglycerides
TNF- α	Tumor necrosis factor-alpha

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12933-026-03096-1>.

Supplementary Material 1.

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None.

Author contributions

H.F.Z., Y.W. and N.W. conceived and designed the study. Y.W. and M.L. performed data acquisition, analysis, and interpretation. Y.W., N.W., and M.L. drafted the manuscript. J.W.L. conducted the statistical analysis. H.F.Z. and H.B.L. provided administrative, technical, and material support. H.F.Z. supervised the study.

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Data availability

The dataset was derived from the China Health and Retirement Longitudinal Study (CHARLS), a publicly accessible resource that can be retrieved by visiting its official website (CHARLS: <https://charls.pku.edu.cn/>).

Declarations

Ethics approval and consent to participate

The National Health and Nutrition Examination Survey protocols were approved by the National Center for Health Statistics Ethics Review Board. Written informed consent was obtained from all participants prior to participation.

Competing interests

The authors declare no competing interests.

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